Heron 2.5 ignition coil B primary circuit failure causing cold-start misfire

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Area: bulletins

1 Condition

Affected Heron 2.5 vehicles may set DTC P0352 (Ignition Coil B Primary/Secondary Circuit Malfunction) accompanied by a rough cold-start misfire that clears as the engine warms. Drivers typically report a brief stumble and an illuminated MIL during the first start of the day. Stored freeze-frame data usually shows the fault occurring at low coolant temperature and low engine speed. For the full diagnostic background on this code, see the Heron Ignition Service Manual :: Coil Primary and Secondary Circuit Faults .

2 Cause

The root cause is an intermittent open in the primary winding of ignition coil B, part number IC-2044-A, when the coil body contracts at low temperatures. As the affected IC-2044-A coil cools, micro-fractures in the primary lead separate and interrupt the driver circuit, producing the P0352 set condition. This behavior is application-specific to the Heron 2.5 coil pack; refer to Heron Ignition Service Manual :: Application-Specific Coils for the affected production range. General coil theory is covered under Heron Ignition Service Manual :: Ignition Coils .

3 Repair Procedure

Replace the affected coil B following PROC-IGN-COIL, torquing the hold-down fastener to specification and verifying the connector latch is fully seated. Use only the superseded IC-2044-A coil with the revised primary lead, and confirm the part number on the new unit before installation. Clear DTC P0352 and perform a cold-start verification drive to confirm the misfire does not return. The complete step-by-step is documented in the Heron Ignition Service Manual :: Ignition Coil Replacement Procedure .

4 Related Service: Spark Plug Inspection

Whenever a coil is replaced for misfire, inspect the corresponding spark plug per PROC-SPARK-PLUG for fouling or excessive gap that could contribute to the misfire. Replace any plug outside specification with part number SP-1110, gapped to the published value. Selection and gap data for SP-1110 is listed in Heron Ignition Service Manual :: Spark Plug Selection and Specifications , and the removal and torque steps are in Heron Ignition Service Manual :: Spark Plug Replacement Procedure .